

Guardrails

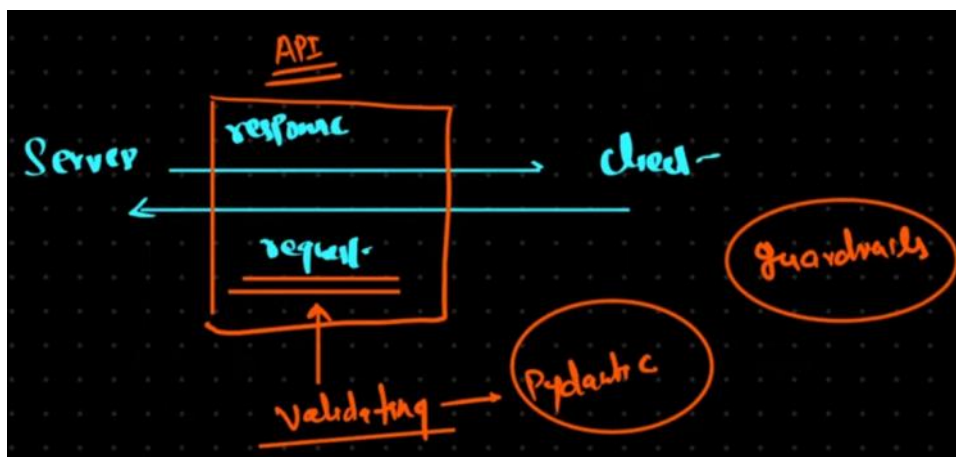
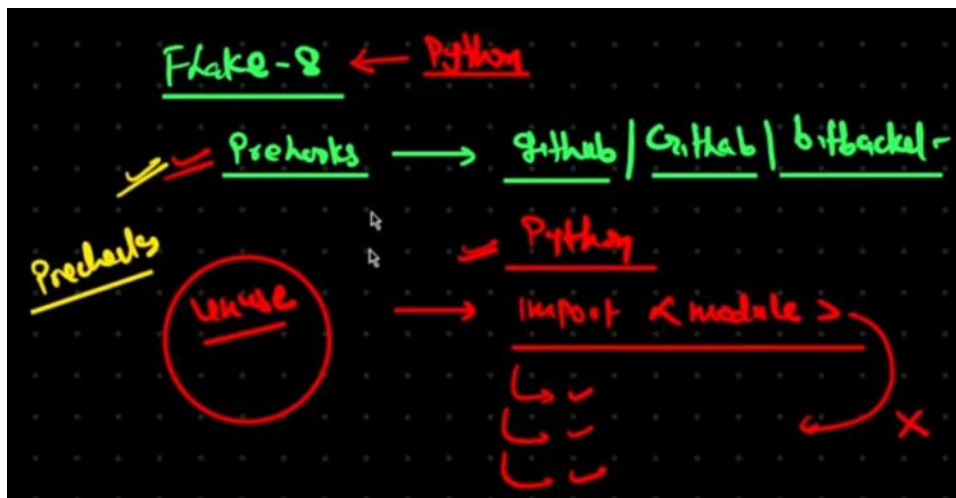
Wednesday, July 29, 2026 7:55 PM

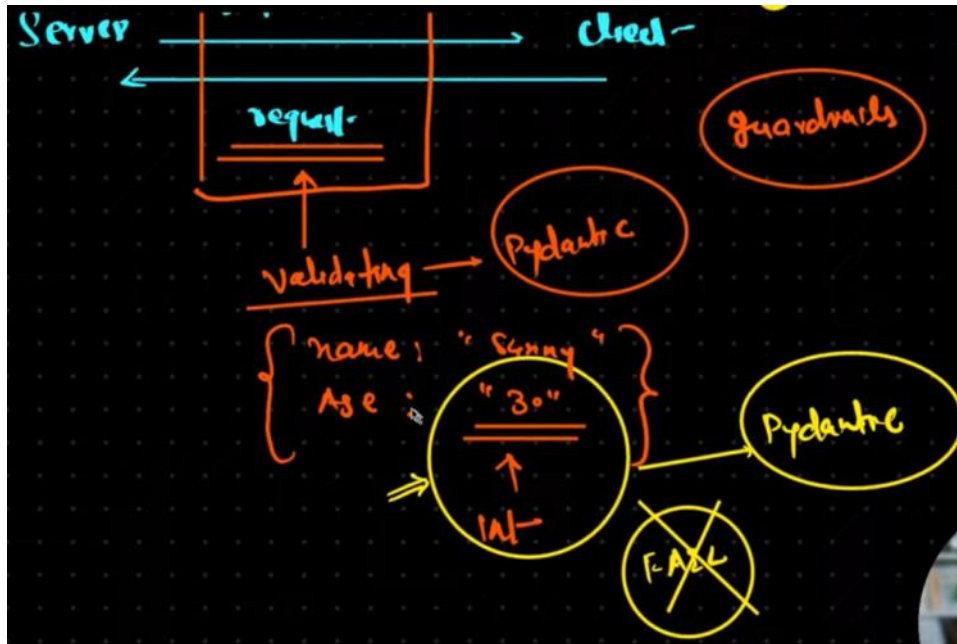
1. What are guardrails?
2. Guardrails in Development world
3. Guardrails in AI -> python library built for llm apps
4. Libraries or frameworks for guardrails
5. Practical with Guardrails AI

Guardrails means Safety boundaries, protective limits, or control mechanisms.

In Software Engineering / Programming / Web Development / DevOps / Cloud & Infrastructure

- **Code linting & Style rules** - Auto-checks enforce coding conventions.
- **Unit & integration tests** - Automated tests ensure new code doesn't break behavior.
- **API validation (e.g: Pydantic)** - Schema-based checks reject malformed input / Output.
- **Form validation** - Client/Server rules block invalid user entries.
- **Role-based access control** - Permission logic limits actions to authorized roles.
- **AWS Service Control Policies (SCP)** - Org-level IAM policies deny unsafe AWS actions.
- **CI/CD checks** - Pipeline gates stop builds or deploys that fail tests / policies.
- **Cost guardrails** - Budget alerts / limits flag or halt unexpected cloud spend.





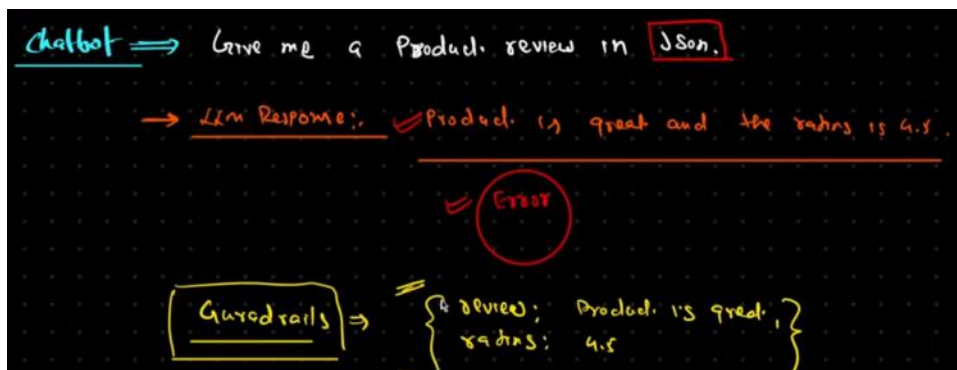
Guardrails in AI

Guardrails in AI (LLM Applications) are the safety boundaries protective limits and control mechanisms that keep a model's behavior on the right, safe, and controlled path.

Definition: Guardrails are rules + safety nets that ensure LLMs stay safe, correct, consistent, and under your control just like a seatbelt keeps you safe in a fast car.

They ensure:

- Correctness (data types, structure)
- Consistency (Output always in expected format)
- Controllability (you define the boundaries, LLM follows)
- Safety (no toxic, offensive, PII leakage)



Medical bot :-

User question: I have chest pain suggest me medicine.

LLM response: take Aspirin and lie down on bed.

Handwritten notes:
 - "chest pain" is boxed in red.
 - "Aspirin" is underlined in red.
 - "not safe response" is written in red with an arrow pointing to the response.
 - "Patient condition" is written in red with an arrow pointing to the response.

After implement the guard rails

take this medicine

LLM response: I am not a medical professional. Please consult with a doctor.

Handwritten notes:
 - "safe" is circled in green.
 - "I am not a medical professional" is underlined in green.

Pythonic libraries:

Guardrail.ai - <https://guardrailsai.com/guardrails/docs>

OpenAI guardrail - <https://guardrails.openai.com/>

NeMo Guardrails - <https://github.com/NVIDIA-NeMo/Guardrails>

LMQL (Language Model Query Language) -

https://runawayhorse001.github.io/GenAI_Best_Practices/html/guardrails.html#

<https://huggingface.co/meta-llama/Llama-Guard-3-8B>

Jailbreak concept:

